MIL-P-46661C (PA)
10 March 1976
SUPERSEDING
MIL-P-46661B (MU)
24 June 1965

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M90
PARTS FOR

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available by all Deparments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts for one type of Percussion Primer designated as Percussion Primer M90.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	- Ammunition and special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1168	- Ammunition Lot Numbering
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Component
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1390

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DRAWINGS

ARMY

8838012 - Cup, Battery
8838015 - Plug, Firing
8866662 - Head
8866665 - Body

(Copies of specifications, standard, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specification.

3.2 Parts.-The parts shall comply with all requirements specified on Drawing (dwg.) 8866662, 8866665, 8838012, 8838015, as applicable, and with all requirements specified in applicable specifications.

3.3 Hydrostatic Test.-The tubing to be fabricated into primer bodies shall withstand the application of a minimum (min.) hydrostatic pressure without rupture or indication of flaws, as evidenced by leaks and without permanent deformation beyond drawing dimensions when tested as specified in 4.4.1 (see 6.3).

3.4 Workmanship.-All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

3.5 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements

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as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 25 each individual components, which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirement of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes, and tests. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample or portion thereof as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspection to be performed.-Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of assembly or

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component in the sample to comply with any or all of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection Provisions

4.3.1 Inspection Lot Formation.-The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. Lot numbering, as required. shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Sampling plans shall be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Plug, firing (see dwg. 8838015)

Categories	Defects	Method of Inspection
Critical:	None defined.	

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Major:

101.	Large diameter, minimum (min.)	Gage
102.	Total length, min.	Gage
103.	Length and profile of point	Gage
104.	Length of pilot, min.	Gage

Minor

201.	Large diameter, maximum (max.)	Gage
202.	Length of pilot, max.	Gage
203.	Diameter of pilot	Gage
204.	Evidence of poor workmanship (see 3.4)	Visual

4.3.2.2 Head (see dwg. 8866662)

Categories	Defects	Method of Inspection

Critical: None defined.

Major:

101.	Major diameter of thread (min.)	Gage
102.	Pitch diameter of thread (min.)	Gage
103.	Concentricity of primer element cavity with body diameter	Gage
104.	Concentricity of flange diameter with body diameter	Gage
105.	Concentricity of flash hole with primer element cavity	Gage
106.	Diameter of primer element cavity, max.	Gage
107.	Depth to bottom of primer element cavity	Gage
108.	Depth of battery cup counterbore, max.	Gage
109.	Diameter of flash hole	Gage
110.	Diameter of body	Gage
111.	Diameter of battery cup cavity	Gage
112.	Chamfer missing on battery cup cavity	Gage
113.	Flash hole missing or obstructed	Visual

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Minor:

201.	Major diameter of thread, max.	Gage
202.	Pitch diameter of thread, max.	Gage
203.	Outside diameter at front end, min.	Gage
204.	Depth of forward cavity	Gage
205.	Length of thread undercut, max.	Gage
206.	Length of shoulder	Gage
207.	Total length, min.	Gage
208.	Thickness of flange, min.	Gage
209.	Diameter of flange, min.	Gage
210.	Diameter of forward cavity	Gage
211.	Length of body threads, including thread undercut	Gage
212.	Finish improper	Visual
213.	Diameter of thread undercut, min.	Gage
214.	Evidence of poor workmanship (see 3.4)	Visual

4.3.2.3 Body (see dwg. 8866665)

Categroies	Defects	Method of Inspection
Critical:		
1.	One or more flash holes missing	Gage
2.	Crack or split	Visual
Major:		
101.	Pitch diameter of head thread (max.)	Gage
102.	Minor diameter of head thread (max.)	Gage
103.	Pitch diameter of closing plug thread (max.)	Gage
104.	Minor diameter of closing plug thread (max.)	Gage
105.	Outside diameter (max.)	Gage
106.	Wall thickness (min.)	Gage
107.	Diameter of holes	Gage
108.	Location of holes	Gage
109.	Depth to which plug gage must enter	Gage

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Minor:

201.	Pitch diameter of head thread (min.)	Gage
202.	Pitch diameter of closing plug thread, min.	Gage
203.	Depth of counterbore, max.	Gage
204.	Diameter of counterbore	Gage
205.	Total length (max.)	Gage
206.	Length of head thread, (min.)	Gage
207.	Length of closing plug thread, min.	Gage
208.	Protective finish with bare spot	Visual
209.	Evidence of poor workmanship (see 3.4)	Visual

4.3.2.4 Battery cup (see dwg. 8838012)

Categories	Defects	Method of Inspection

Critical: None defined.

Major:

101.	Outside diameter, min.	Gage
102.	Thickness through bottom, min.	Gage
103.	Total length	Gage

Minor:

201.	Diameter of hole, max.	Gage
202.	Thickness through bottom, min.	Gage
203.	Inside diameter of cavity	Gage
204.	Outside diameter, max.	Gage
205.	Flat missing from surface at open end	Visual
206.	Evidence of poor workmanship (see 3.4)	Visual

4.3.3 Testing

4.3.3.1 Hydrostatic Test (see 3.3)- Critical defect.- This test shall be conducted 100 percent. If any primer tubing fails to comply with the applicable requirement, it shall be classed defective and no primer bodies shall be fabricated therefrom. The test shall be performed as specified in 4.4.1 using equipment specified in 4.3.4.

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4.3.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.3.2 and 4.3.3 (see 6.2).

4.4 Test Methods and Procedures

4.4.1 Hydrostatic Test.-Prior to machining, the primer tubing shall be tested for not less than five seconds at the min. hydrostatic pressure as calculated by the following formula:

P = 1.8 St divided by D

where:
P = test pressure in pounds per square inch (p.s.i.)
S = min. specified yield strength, p.s.i.
t = specified min. wall thickness in inches.
D = specified max. outside diameter in inches.

Observation shall be made for rupture, or indication of flaws as evidenced by leaks, and for permanent deformation beyond drawing requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.-The parts shall be packaged, packed and marked in accordance with Standard MIL-STD-1169.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. First article inspection provisions.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are

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designated by drawing numbers under the "Method of Inspection" heading in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE Designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor Designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204 "Inspection Equipment Design" for guidance. The item detail specification number, paragraph number, and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter and date to which the specific design applies.

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6.2.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Designs shall be submitted for approval to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

6.3 Hydrostatic Test.-If the primer tubing procured for the bodies manufactured in accordance with this specification has been previously subjected to and has successfully passed the hydrostatic test specified in this specification, a second hydrostatic test will not be required.

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Preparing Activity:
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Project Number: 1390-A105

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